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Origin of the Different pH Activity Profile in Two Homologous Ketosteroid Isomerases*

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Two homologous Δ^5-3-ketosteroid isomerases from *Comamonas testosteroni* (TI-WT) and *Pseudomonas putida* biotype B (PI-WT) exhibit different pH activity profiles. TI-WT loses activity below pH 5.0 due to the protonation of the conserved catalytic base, Asp-38, while PI-WT does not. Based on the structural analysis of PI-WT, the critical catalytic base, Asp-38, was found to form a hydrogen bond with the indole ring NH of Trp-116, which is homologously replaced with Phe-116 in TI-WT. To investigate the role of Trp-116, we prepared the F116W mutant of TI-WT (TI-F116W) and the W116F mutant of PI-WT (PI-W116F) and compared kinetic parameters of those mutants at different pH levels. PI-W116F exhibited significantly decreased catalytic activity at acidic pH like TI-WT, whereas TI-F116W maintained catalytic activity at acidic pH like PI-WT and increased the k_{cat}/K_m value by 2.5- to 4.7-fold compared with TI-WT at pH 3.8. The crystal structure of TI-F116W clearly showed that the indole ring NH of Trp-116 could form a hydrogen bond with the carboxyl oxygen of Asp-38 like that of PI-WT. The present results demonstrate that the activities of both PI-WT and TI-F116W at low pH were maintained by a tryptophan, which was able not only to lower the pK_a value of the catalytic base but also to increase the substrate affinity. This is one example of the strategy nature can adopt to evolve the diversity of the catalytic function in the enzymes. Our results provide insight into deciphering the molecular evolution of the enzyme and creating novel enzymes by protein engineering.

Isozymes carry out the same enzymatic reaction yet exhibit different properties such as substrate affinity, preference of different coenzyme, stability, optimal pH, etc. Even if the sequence differences of those isozymes are evident, the origin of different properties is mostly not clear at the molecular level. Identification of functional divergence on the mechanistic basis is important to understand how the enzyme as a molecular machine can be evolved in the biological system and to rationally modify the enzyme function with desired physical and catalytic properties (1). In the post-genomic era, protein sequences and high resolution protein structures are being accumulated rapidly, but the remarkable catalytic mechanism of enzymes as well as the catalytic strategies utilized by nature to evolve new catalytic functions remain to be understood at the molecular level (2). The understanding of the structural basis for different properties of isozymes and their catalytic diversity could provide the insights into not only discovering the properties of the enzymes from their protein sequences or structural data but also creating novel enzymes by protein engineering.

Δ^5-3-Ketosteroid isomerase (KSI,[1] EC 5.3.3.1) is one of the most proficient enzymes catalyzing an allylic isomerization reaction at a rate comparable to the diffusion-controlled limit by an intramolecular transfer of a proton (Scheme 1) (3–8). Although KSIs from two different bacterial sources, *Comamonas testosteroni* and *Pseudomonas putida* biotype B, share only 34% sequence identity, the catalytic residues Tyr-14, Asp-38, and Asp-99 (the residues are numbered according to *C. testosteroni* KSI) are well conserved in the two isomerases (9, 10). Moreover, the overall three-dimensional structures are remarkably similar to each other, indicating that the two KSIs can share the same catalytic mechanism (11–14). In KSIs, Asp-38 acts as the only catalytic base to abstract a proton from C-4 of the steroid substrate and to transfer the proton to the C-6 position for the allylic rearrangement of Δ^5-3-ketosteroids (3, 14).

KSI from *P. putida* (PI-WT) was reported to exhibit different dependence of V_{max} and K_m on pH compared with that from *C. testosteroni* (TI-WT) (15). The V_{max} value of PI-WT was not so sensitive to pH as that of TI-WT, which had been observed before (15, 16). The pH-rate profiles for k_{cat} and k_{cat}/K_m were extensively investigated for TI-WT (17). The pK_a of the catalytically important group on the free enzyme (pK_E) was determined to be 4.57 utilizing a nonsticky substrate, 5,10-estrene-3,17-dione (17). The catalytic activity of TI-WT is supposed to decrease abruptly below pH 5.0 due to the protonation of the catalytic base, Asp-38. The pK_a values of catalytic residues can be affected by the local microenvironment of the active site in the enzyme (18). The environment of Asp-38 in TI-WT is very

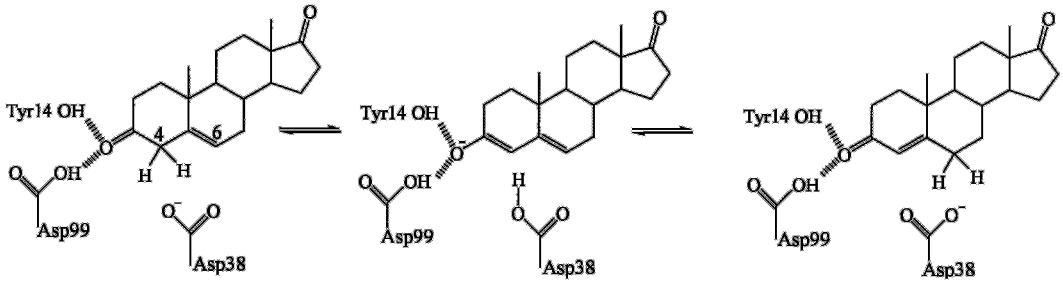
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The atomic coordinates and structure factors (code 1OCV) have been deposited in the Protein Data Bank, Research Collaboratory for Structural Bioinformatics, Rutgers University, New Brunswick, NJ (http://www.rcsb.org/).

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[1] The abbreviations used are: KSI, Δ^5-3-ketosteroid isomerase (EC 5.3.3.1); 5-AND, 5-androstene-3,17-dione; 5,10-EST, 5,10-estrene-3,17-dione; CD, circular dichroism; WT, wild-type; MES, 2-(*N*-morpholino)-ethanesulfonic acid; PI-WT, wild-type KSI from *P. putida* biotype B; PI-W116F, W116F mutant of PI-WT; TI-WT, wild-type KSI from *C. testosteroni*; TI-F116W, F116W mutant of TI-WT; K_E, ionization constant of the free enzyme; K_{ES}, ionization constant of the enzyme-substrate complex.

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SCHEME 1. **Catalytic mechanism of KSI.** The reaction has been known to proceed via a dienolate intermediate by a concerted action of three catalytic residues, Tyr-14, Asp-38, and Asp-99.

similar to that in PI-WT as judged by crystallographic data of both wild-type enzymes (11, 13, 19). However, there is a significant difference in that the carboxyl oxygen of Asp-38 is hydrogen-bonded to NH on the indole ring of Trp-116 only in PI-WT (Fig. 1*B*). Trp-116 in PI-WT is homologously replaced with Phe-116 in TI-WT (Fig. 1*C*). The different electrostatic environments around the catalytic bases in TI-WT and PI-WT could affect the pK_a of Asp-38, resulting in different pK_E values of two homologous KSIs.

To investigate the role of Trp-116 in PI-WT, we have prepared the F116W mutant of TI-WT (TI-F116W) and the W116F mutant of PI-WT (PI-W116F), and compared their kinetic parameters at various pHs, respectively. The crystal structure of TI-F116W determined at 2.0-Å resolution exhibited that the indole ring NH of Trp-116 could form a hydrogen bond with the carboxyl oxygen of Asp-38. Our kinetic and structural results demonstrate that different pH activity profiles in the two KSIs originate from a single hydrogen bond between Trp-116 and Asp-38 and that Trp-116 also contributes to the better binding of the hydrophobic steroid substrate to compensate for the lowered basicity of the catalytic residue.

EXPERIMENTAL PROCEDURES

Materials—5-Androstene-3,17-dione (5-AND), and 5,10-estrene-3,17-dione (5,10-EST) were purchased from Steraloids. Chemicals for buffer solutions were from Sigma. Oligonucleotides were from Genotech Inc., Korea. A QuikChange site-directed mutagenesis kit was supplied by Stratagene. pKK223-3 plasmid was from Amersham Biosciences. Superose 12 gel filtration column was obtained from Amersham Biosciences.

Mutagenesis—Phe-116 of TI-WT was replaced with a tryptophan to make TI-F116W. TI-F116W was prepared using the QuikChange site-directed mutagenesis kit (Stratagene) and a thermocycler (Minicycler, MJ Research). pKSI-TI was used as the DNA template (11). Two primers were designed to introduce the F116W mutation into the KSI gene of *C. testosteroni*. A forward primer was 5′-AGC ATG CGC GCC TTG TGG GGC GAG AAG AAT ATT-3′ and a reverse primer was 5′-AAT ATT CTT CTC GCC CCA CAA GGC GCG CAT GCT-3′; underlined nucleotides represent the ones changed by point mutations. Recombinant plasmids were introduced into *Escherichia coli* XL1-Blue supercompetent cells (Stratagene) and purified by use of QIAprep Spin Miniprep kit (Qiagen). The entire KSI gene was then sequenced to confirm the intended mutation only without the change of other sequences in the KSI gene. The W116F mutation of PI-WT was described previously (20).

Expression and Purification of KSI Proteins—The protein was overproduced in *E. coli* BL21(DE3) utilizing the pKK223-3 plasmid containing the wild-type or mutant KSI gene and purified by deoxycholate affinity chromatography and Superose 12 gel filtration chromatography as described previously (21). The purity of the protein was confirmed by a single band on SDS-PAGE. The protein concentration was determined utilizing the difference extinction coefficient between tyrosinate and tyrosine at 295 nm as described previously (22). The accuracy of the protein concentration was confirmed by the quantitative analysis of the band on SDS-PAGE by use of an imaging densitometer (Bio-Rad, GS-700) and a software program provided by its manufacturer (Molecular Analyst/PC).

Steady-state Kinetic Analysis—The catalytic activities of the purified KSIs were determined spectrophotometrically using 5-AND as a substrate according to the procedure described previously (21). Various amounts of the substrate were added to a reaction buffer containing 34 mM potassium phosphate and 2.5 mM EDTA, pH 7.0, at 25 °C. The concentration of 5-AND was 12, 35, 58, 82, and 116 μM, respectively. The final concentration of methanol was 3.3 vol%. The initial reaction rate was obtained within 1 or 2 min after the initiation of the enzymatic reaction. The fraction of the substrate converted to the product was below 10% of the substrate applied to the reaction mixture. The reaction was monitored by measuring the absorbance at 248 nm, using a spectrophotometer (Shimadzu, UV-2501 PC). Kinetic data were analyzed by use of Lineweaver-Burk reciprocal plots to obtain k_{cat} and K_m.

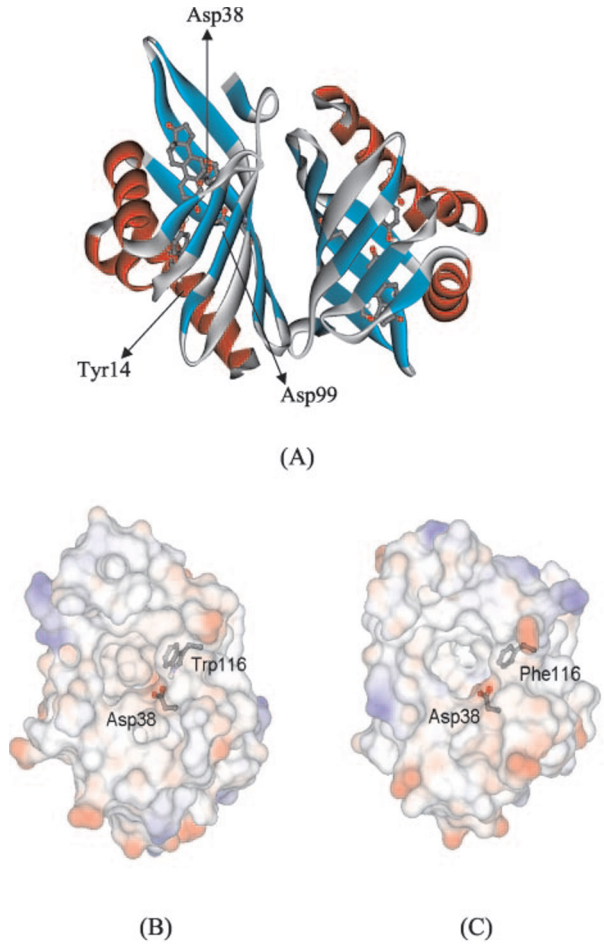


FIG. 1. **Three-dimensional structure of KSI.** *A*, ribbon diagram of a dimeric KSI from *P. putida* KSI complexed with *d*-equilenin, an intermediate analogue. Three catalytic residues and *d*-equilenin are shown in *ball-and-stick* model representation. *B*, monomeric structure of the dimeric KSI from *P. putida*. *C*, monomeric structure of the dimeric KSI from *C. testosteroni*. The molecular surface of each KSI monomer using the probe of 1.4 Å radius was generated. Positive and negative charged regions are indicated by *blue* and *red*, respectively. Asp-38 and Trp-116, or Phe-116 are shown in *ball-and-stick* model representation. The figures were drawn using the programs WebLab ViewerPro; Accelrys.

TABLE I
Kinetic parameters of WT and mutant KSIs

pK_a values for PI-WT, TI-WT, and T1-F116W were determined with a nonsticky substrate, 5,10-estrene-3,17-dione. pK_a values for PI-W116F were determined with 5-androstene-3,17-dione.

Enzyme	k_{cat}	K_M	k_{cat}/K_M	pK_E	pK_{ES}
	S^{-1}	μM	$M^{-1}\ s^{-1}$		
PI-WT	21230 ± 810[a]	49.9 ± 1.3[a]	4.25×10^8	3.75 ± 0.04	4.24 ± 0.15
PI-W116F	4760 ± 36[a]	149 ± 20[a]	3.19×10^7	5.00 ± 0.20	5.25 ± 0.20
TI-WT	31683 ± 1200	152.8 ± 8.2	2.07×10^8	4.57 ± 0.05	4.76 ± 0.04
TI-F116W	7155 ± 194	48.6 ± 1.4	1.47×10^8	3.52 ± 0.04	<3.8[b]

[a] Data from Kim *et al.* (20).
[b] The upper limit was indicated, because k_{cat} did not decrease even at pH 3.8 using the above substrates.

pH Activity Profiles—All the buffers used to determine the kinetic constants contained 80 mM NaCl to adjust the ionic strength of the solution. The buffer solution contained 20 mM sodium acetate for pH 3.8–4.9, 20 mM sodium MES for pH 5.2–6.3, or 20 mM potassium phosphate for pH 6.6–8.0. The procedures for obtaining kinetic parameters were the same as those for the steady-state kinetic analysis as described above. The stability of the enzyme was confirmed by determining the catalytic activity at pH 7.0 after incubating the enzyme in the respective assay buffer for 2 min. The observed kinetic parameters were fitted to Equations 1 and 2 to obtain the pK_E and pK_{ES} values, respectively, by nonlinear least-squares analysis utilizing a computer program (Abelbeck Software, Kaleidagraph version 3.06).

$$(k_{cat}/K_M)_{obs} = (k_{cat}/K_M)/(1 + [H^+]/K_E) \quad \text{(Eq. 1)}$$

$$(k_{cat})_{obs} = (k_{cat})/(1 + [H^+]/K_{ES}) \quad \text{(Eq. 2)}$$

To obtain the pK_E and pK_{ES} values of TI-WT, TI-F116W, and PI-WT from the pH activity profile, a nonsticky substrate, 5,10-EST, was used as described previously (17). Similarly, the pK_E and pK_{ES} values of PI-W116F were obtained using 5-AND, which can be considered a nonsticky substrate, because its k_{cat}/K_m values are at least one order of magnitude smaller than that for PI-WT as described previously (23). To compare the kinetic parameters of TI-WT and TI-F116W, both 5,10-EST and 5-AND were used. The plot of $(k_{cat}/K_m)_{obs}$ *versus* pH was utilized to determine pK_a for the free enzymes (pK_E) and the plot of $(k_{cat})_{obs}$ *versus* pH to determine pK_a for the enzyme-substrate complexes (pK_{ES}) (18, 27, 28).

Crystallization and Structure Determination—The crystal of TI-F116W was grown at 25 °C from a solution containing 4% polyethylene glycol 400, 1.6 M ammonium sulfate, and 0.1 M HEPES, pH 7.5, by the hanging-drop method as described previously (11). The crystal was briefly immersed in the same solution containing 15% glycerol. Then the crystal was flash-cooled in the stream of cooled nitrogen gas (100 K) prior to x-ray data collection. Diffraction data were obtained by using an R-AXIS IV^{++} image plate with a rotating anode generator (Rigaku, RU300). The crystal belongs to the space group P3 (1) with unit cell dimensions of $a = 71.525$, $b = 71.525$, $c = 103.340$ Å, and $\gamma = 120°$. Data reduction, merging, and scaling were carried out by use of the programs DENZO and SCALEPACK, as described previously (25). The structure was determined by using a molecular replacement program, AMORE, utilizing the atomic coordinates of TI-WT (PDB code 8CHO). Further refinement was carried out by use of the program CNS as described previously (26).

Circular Dichroism Spectroscopic Analysis—Circular dichroism (CD) spectra of the enzymes were obtained with a spectropolarimeter (Jasco, 715) equipped with a Peltier-type temperature controller (Jasco, PTC-348WI), using a quartz cuvette with 0.2-cm path length. The wild-type and mutant enzymes at 10 μM were preincubated at 25 °C in a buffer containing 20 mM sodium acetate for pH 3.8, 4.2, and 5.0, 20 mM sodium MES for pH 6.0, or 20 mM potassium phosphate for pH 7.2. Scans were collected at 0.1-nm intervals with 1-nm bandwidth and accumulated three to five times. Each spectrum was corrected by subtracting the spectrum of the solution containing the used buffer and smoothened by use of a software program provided by the manufacturer of the spectropolarimeter.

Native Polyacrylamide Gel Electrophoresis—Native PAGE was carried out as described previously with a slight modification (23, 24). The wild-type and mutant enzymes were subjected to electrophoresis in 8% polyacrylamide gels in 50 mM sodium acetate, pH 4.3, or 50 mM HEPES, pH 7.3. Electrophoresis was carried out at 4 °C to keep the gel from being overheated by high voltage. During the electrophoresis, the tray buffer was circulated between two reservoirs to maintain constant pH. Each gel was pre-run at 150 V for 1 h, after which 1.0–2.0 μg of the protein in the respective buffer solution containing 10% glycerol and 0.003% bromphenol blue was loaded on the gel and run at 100 V.

RESULTS

Kinetic Parameters—k_{cat} and K_m values of TI-F116W and PI-W116F were compared with those of their respective wild-type enzyme (Table I). The k_{cat} values of TI-F116W and PI-W116F were lower than those of TI-WT and PI-WT by 4.42- and 4.46-fold, respectively. The K_m value of PI-W116F was higher than that of PI-WT by 2.98-fold, whereas the K_m value of TI-F116W was lower than that of TI-WT by 3.14-fold. TI-F116W exhibited the k_{cat}/K_m value nearly comparable to that of the wild-type, because both K_m and k_{cat} values were decreased simultaneously at pH 7.0.

pH Activity Profiles—To estimate the pK_a values of the catalytic base in two KSIs, kinetic parameters were determined at various pHs (Fig. 2). The observed kinetic parameters, which were fitted to Equations 1 and 2 to obtain the pK_E and pK_{ES} values, respectively, are listed in Table I. The pK_E values of TI-WT were consistent with the previous results for two substrates, 5-AND and 5,10-EST (17). The pK_E value in TI-F116W was decreased by 1.05 pH unit relative to that of TI-WT. At pH 3.8, the k_{cat}/K_m value of TI-F116W was increased by 2.5-fold and by 4.7-fold for two substrates, 5,10-EST and 5-AND, respectively, compared with that of TI-WT. The apparent pK_a values of Asp-38 in PI-WT and in PI-W116F were estimated to be 3.75 and 5.00, respectively, with a significant decrease of catalytic activity at low pH in PI-W116F (Fig. 2).

Structural Analysis of TI-F116W—To identify a hydrogen bond formed by the indole ring NH of Trp-116 in TI-F116W, the crystal structure of TI-F116W was determined at 2.0-Å resolution. The statistics of the crystallographic data and refinement are summarized in Table II. No significant structural changes were observed in the mutant KSI compared with TI-WT. The structure of TI-F116W exhibited that the carboxyl oxygen of Asp-38 could form a hydrogen bond with the indole ring NH of Trp-116, and the distance between Asp-38 Oδ1 and Trp-116 Nϵ1 was measured to be 2.87 Å. Structural comparison of PI-WT, TI-WT, and TI-F116W indicated clearly that the hydrogen bond formed by the indole ring NH of tryptophan with the carboxyl oxygen of the catalytic base was the only difference of the environment around Asp-38 between two wild-type KSIs, and between TI-WT and TI-F116W (Fig. 3). For simulated annealing omit map, the structure was disturbed at 1000 K at an initial refinement stage when the residues, Trp-116 and Asp-38, were omitted. The electron density map was then calculated with the omission. Trp-116 and Asp-38 exhibited clear electron densities, supporting that the indole ring NH of Trp-116 can form a hydrogen bond with the carboxyl group of Asp-38 (Fig. 4).

CD Spectra at Different pHs—To investigate the effect of pH on the conformation of KSIs, far-UV CD spectra were obtained in the pH range from 3.8 to 7.2 (Fig. 5). CD spectra were obtained between 200 and 250 nm to estimate the content of the secondary structure in the protein. TI-WT and TI-F116W

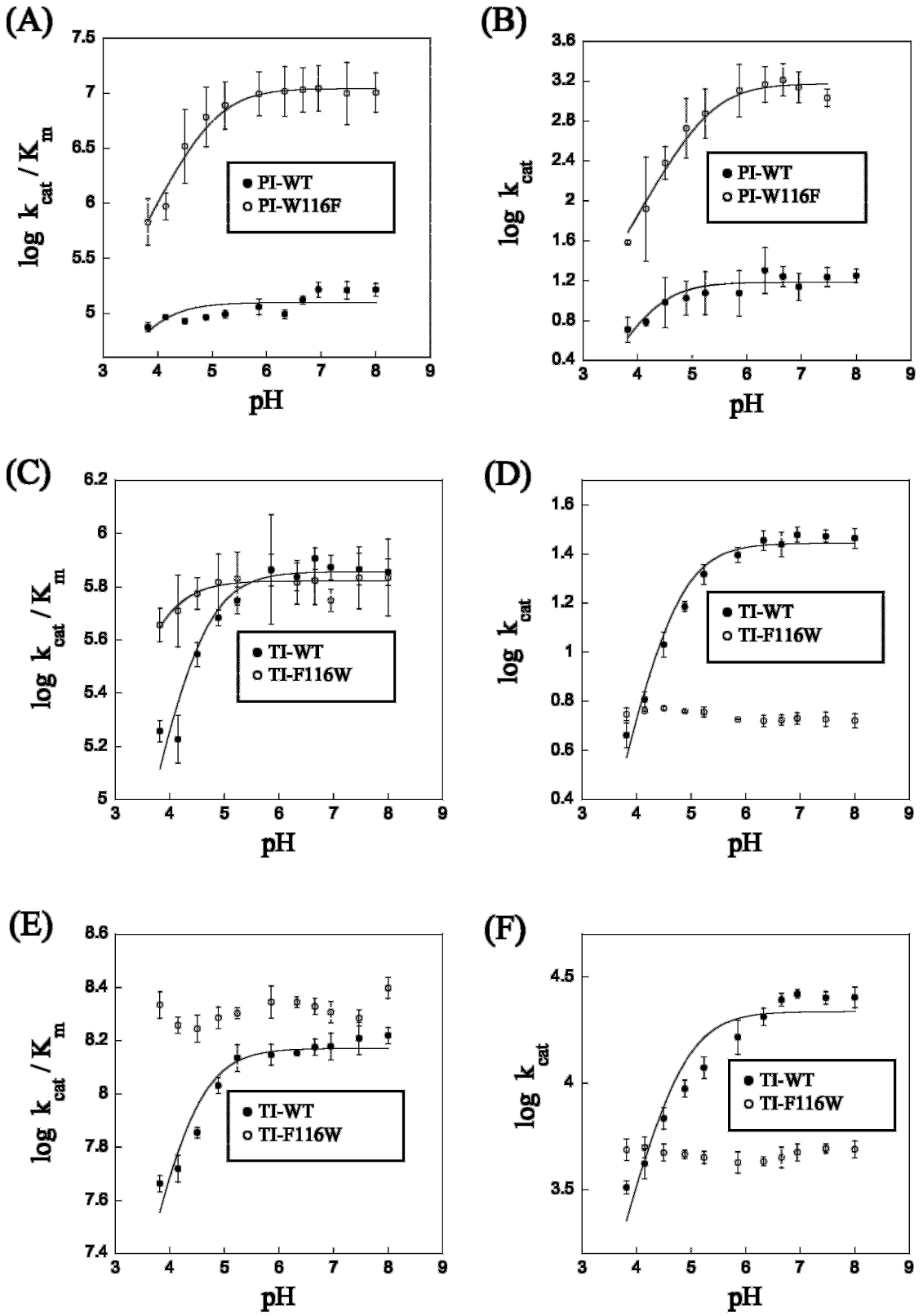


FIG. 2. **pH dependence of kinetic parameters of WT and mutant KSIs.** The *lines* represent nonlinear least-squares fits of the data to Equations 1 and 2 to obtain the pK_E and pK_{ES} values, respectively, as listed in Table I. 5,10-EST was used as a substrate for *graphs C* and *D*, and for PI-WT in *graphs A* and *B*. 5-AND was used as a substrate for *graphs E* and *F*, and for PI-W116F in *graphs A* and *B*.

exhibited similar spectra in the pH range from 3.8 to 7.2, respectively, whereas the spectra of PI-WT and PI-W116F were marginally changed below pH 5.0, indicating that there were no significant structural changes over the pH range from 3.8 to 7.2 in all different forms of KSI.

Native PAGE—The ionization state of the catalytic base in the two KSIs can affect the overall charge of the enzyme, because the catalytic base, Asp-38, is located in the boundary between the cavity of the hydrophobic active site and the solvent (Fig. 1). Mobilities of PI-WT, PI-W116F, TI-WT, and TI-

F116W in the gel electrophoresis were compared at two different pHs, 4.3 and 7.3 (Fig. 6). PI-WT and PI-W116F had the same mobility at pH 7.3, whereas PI-W116F migrated more slowly than PI-WT toward the anode at pH 4.3. TI-WT and TI-F116W also had the same mobility at pH 7.3, whereas TI-WT migrated more slowly than TI-F116W toward the anode at pH 4.3, indicating that TI-WT and PI-W116F have less effective negative charges than TI-F116W and PI-WT, because the carboxylic group of Asp-38 in those KSIs could be protonated at pH 4.3.

TABLE II
Crystallographic data and refinement statistics for TI-F116W

Resolution (Å)	2.0
R_{sym} (%)	6.0
Data completeness, $F > 1\ \sigma$ (%)	91.8
$R_{standard}$ (%)	22.88
R_{free} (%)	28.24
No. of refined atoms (atom/water)	4161/377
Average *B* factor	21.454
r.m.s.d. bond length (Å)[a]	0.005970
r.m.s.d. bond angles (deg)	1.16570
Ramachandran plot (%)	
Most favored regions	90.8
Additional allowed regions	8.3
Generously allowed regions	0.9
Disallowed regions	0.0

[a] r.m.s.d., root mean square deviation.

DISCUSSION

The analyses of pH activity profiles and crystallographic structures at high resolution for the wild-type and mutant forms of two homologous KSIs from two different bacterial sources allowed us to identify the role of Trp-116, including a hydrogen bond formed by the indole ring NH with the catalytic base, Asp-38, in PI-WT. A different environment around Asp-38 was found to be the origin of the different pH activity profiles of the two KSIs. The carboxylate ion could be destabilized in a relatively hydrophobic region of the active sites in TI-WT and PI-W116F, resulting in the increase of the pK_a value

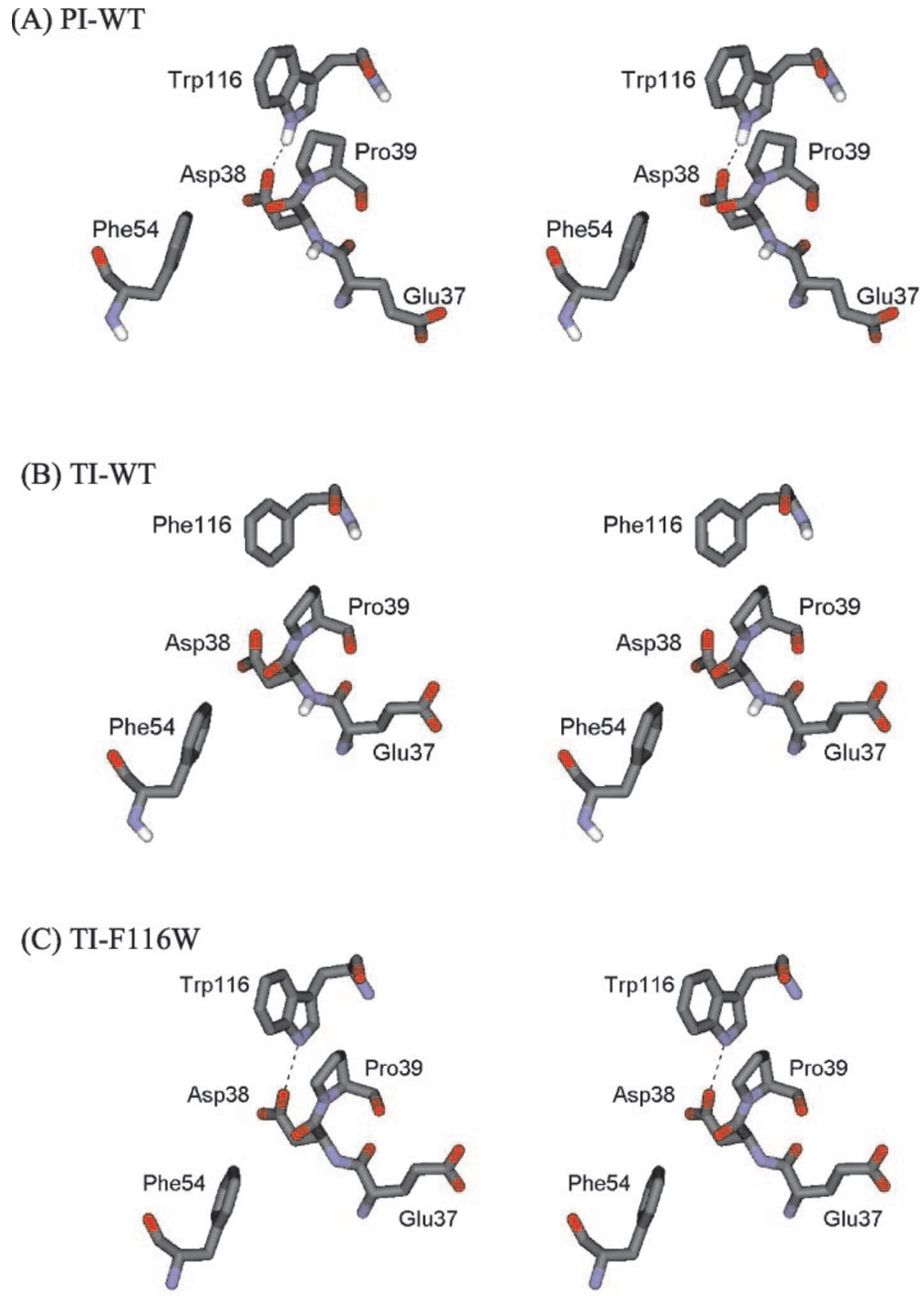


FIG. 3. **Stereoview of the catalytic bases of PI-WT (*A*), TI-WT (*B*), and TI-F116W (*C*).** Hydrogen bonds between Asp-38 and Trp-116 are represented by *dashed lines*. The distance between Asp-38 Oδ1 and Trp-116 Nϵ1 in PI-WT was 2.87 Å (*A*). Phe-116 in TI-WT was located at the corresponding position of Trp-116 in PI-WT (*B*). The distance between Asp-38 Oδ1 and Trp-116 Nϵ1 in TI-F116W was 2.87 Å (*C*). The figures were drawn using the programs WebLab ViewerPro; Accelrys. PDB codes are 4TSU and 8CHO for PI-WT and TI-WT, respectively.

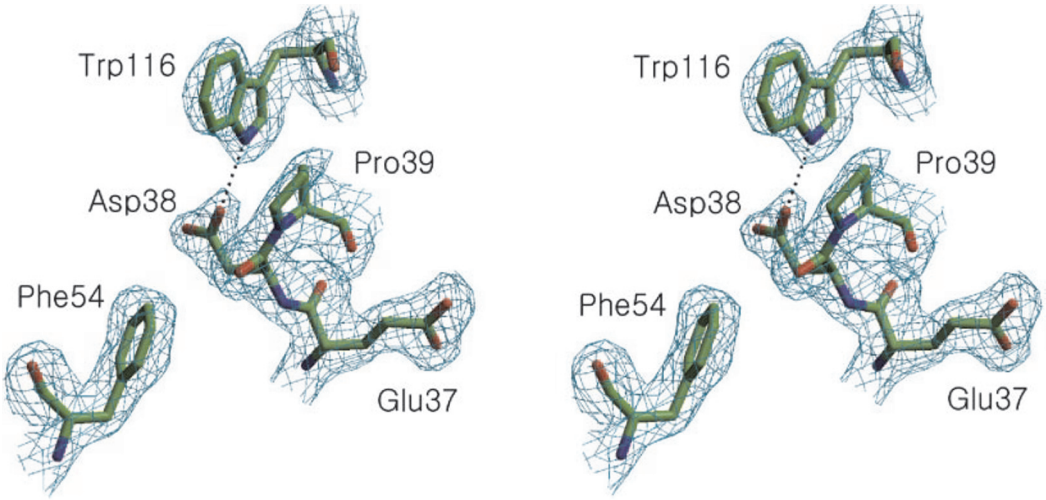


FIG. 4. **Stereoview of the catalytic base of TI-F116W with $2F_o - F_c$-simulated annealing omit electron density map contoured at 1.0 σ.** Residues Trp-116 and Asp-38, which were omitted from the model, display clear electron density. A hydrogen bond between Asp-38 and Trp-116 is represented by a *dashed line*. The figure was drawn by using the program BobScript and rendered using Raster3D.

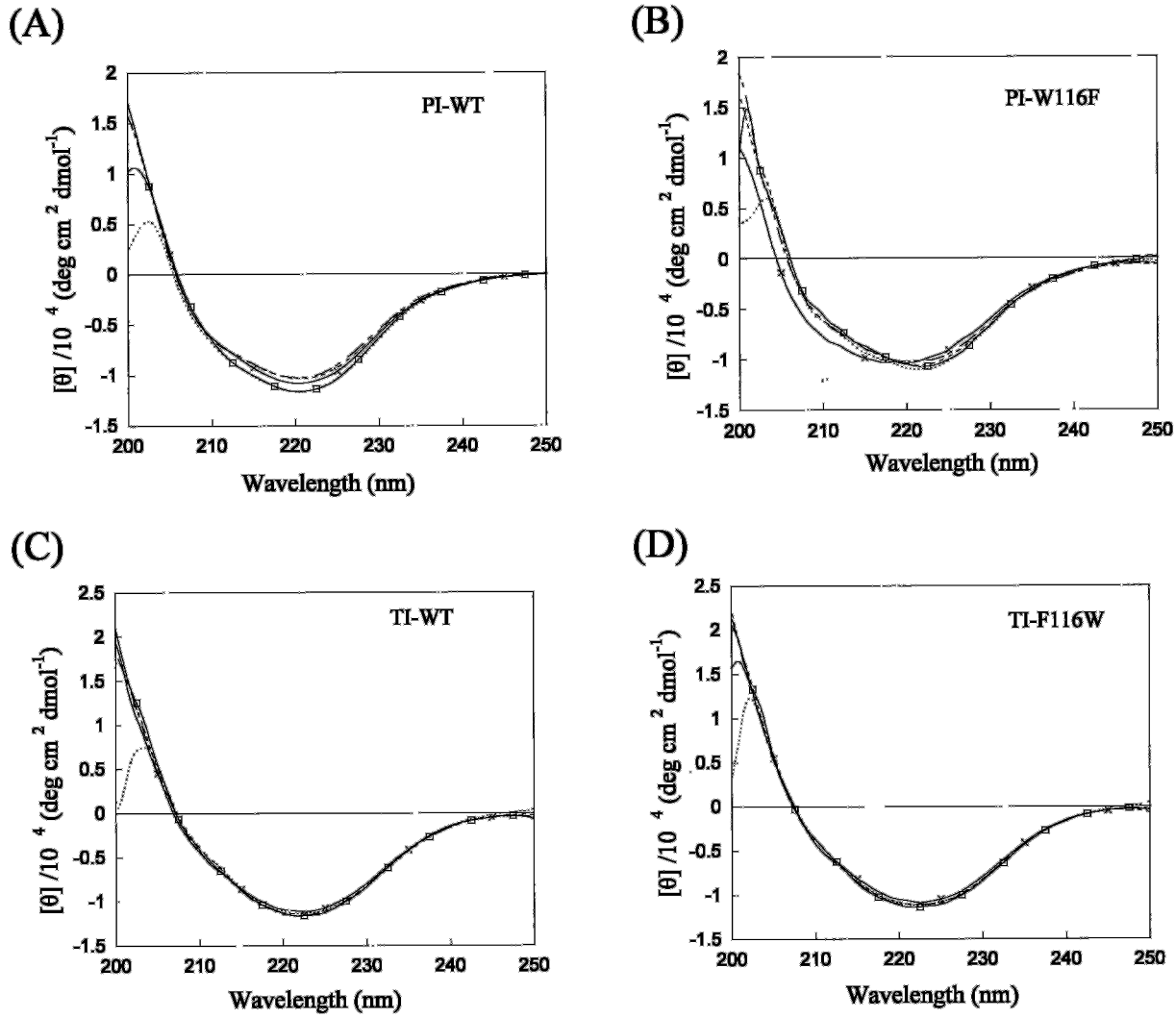


FIG. 5. **Far-UV CD spectra of WT and mutant KSIs at different pH.** Spectra of 10 μM KSI were obtained in the buffer solution containing 20 mM sodium acetate for pH 3.8, 4.2, and 5.0, 20 mM sodium MES for pH 6.0, or 20 mM potassium phosphate for pH 7.2.; pH 3.8 (×), pH 4.2 (*solid line*), pH 5.0 (*dashed line*), pH 6.0 (*dotted line*), and pH 7.2 (□).

of Asp-38. In contrast, the pK_a values of Asp-38 in PI-WT and TI-F116W could be decreased, because the carboxylate ion in Asp-38 was stabilized by a hydrogen bond with Trp-116 (Fig. 7).

Considering the different environment of the critical catalytic base, Asp-38, in two KSIs, the low pK_E value of PI-WT can be attributed to a hydrogen bond between the carboxylate of

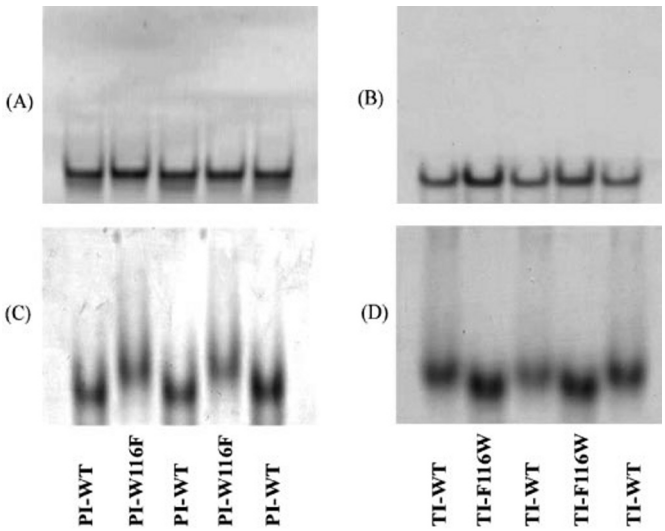


FIG. 6. **Native PAGE of WT and mutant KSIs.** The direction of migration is toward the anode at the *bottom* of the gels at pH 7.3 in *A* and *B*, and at pH 4.3 in *C* and *D*. The same samples were loaded in different lanes as indicated to compare the migration between the WT and mutant KSIs more precisely.

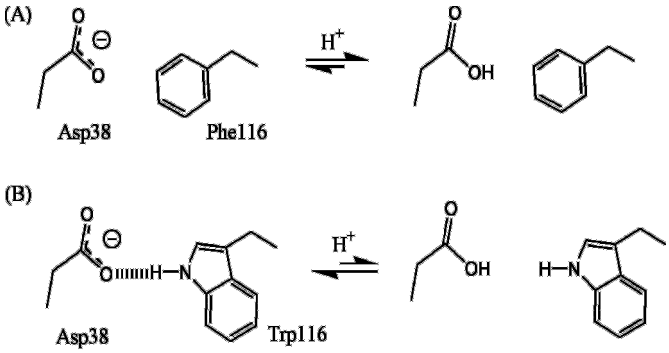


FIG. 7. **Putative ionization states of the catalytic base, Asp-38, in the WT and mutant KSIs at pH 4.0.** The pK_a values of Asp-38 in TI-WT and PI-W116F (*A*) can be higher than those of Asp-38 in PI-WT and TI-F116W (*B*).

Asp-38 and the indole ring NH of Trp-116. The replacement of Trp-116 with a phenylalanine in PI-WT to mimic the active-site environment of TI-WT resulted in a significant decrease of the catalytic activity at low pH, and the W116F mutation increased the pK_E value by 1.25 relative to that of PI-WT. The increase of pK_E in PI-W116F is due to the pK_a increase of Asp-38, because the hydrogen bond between Trp-116 and the carboxylate of Asp-38 could not be formed. The low mobility of PI-W116F on native PAGE is the evidence for the protonated state of Asp-38 at pH 4.3, because the apparent pK_a value of Asp-38 was kinetically estimated to be 5.0 in PI-W116F. The far-UV CD spectra of PI-WT and PI-W116F between 200 and 250 nm were almost indistinguishable at low pH, indicating that the enzymes did not undergo any significant conformational change at low pH. Therefore, the decreased activity of PI-W116F at low pH originates not from any structural change of the protein, but from the protonation of the catalytic base.

TI-F116W was prepared to confirm that a hydrogen bond between Trp-116 and Asp-38 could compensate for the decreased activity of TI-WT at low pH by stabilizing the ionization state of the catalytic base, Asp-38. As expected, TI-F116W exhibited no decrease of catalytic activity at low pH and increase of k_{cat}/K_m by 4.7- and 2.5-fold compared with TI-WT when 5-AND and 5,10-EST were used as substrates at pH 3.8, respectively. As shown in the crystal structure of TI-F116W, a hydrogen bond can be formed by the indole ring NH of Trp-116 with a carboxyl oxygen of Asp-38 (Fig. 3). The hydrogen bond could cause the pK_a value of Asp-38 to be decreased in TI-F116W. The lower mobility of TI-WT relative to TI-F116W on native PAGE at pH 4.3 was also consistent with the pK_a values estimated kinetically, reflecting the change of ionization state of the critical catalytic base, Asp-38. The far-UV CD spectra of TI-WT and TI-F116W showed almost identical patterns in the pH range from 3.8 to 7.2, indicating that any significant conformational change over the pH range did not take place. Hence, the low activity of TI-WT at low pH is not from the structural change but from the protonation of the catalytic base, Asp-38.

The importance of the indole ring NH of tryptophan was previously suggested in its interaction with other residues in the protein (29, 30). Tryptophan can have characteristics of both hydrophobicity and hydrophilicity due to the amphipathic property, which originates from the indole ring in its side chain (30). The amphipathic characteristic of tryptophan plays two different roles in the catalysis of KSIs. One role of tryptophan is that the bulky hydrophobic indole ring contributes to the better binding of the hydrophobic substrate like steroids. Compared with TI-WT and PI-W116F, PI-WT and TI-F116W have relatively lower K_m values for the hydrophobic steroid substrate (Table I). This notion that Trp-116 contributes to the favorable binding of the steroid is consistent with the previous result in the characterization of PI-W116F (20). The other role of tryptophan is the stabilization of ionized state of the catalytic base residue, Asp-38, at low pH in the hydrophobic active site of KSI, due to a hydrogen bond between Trp-116 and Asp-38. Although the hydrogen bond could decrease the basicity of Asp-38, this adverse effect for catalytic activity could be overcome by the increased binding affinity to the substrate.

The alteration of pH activity profiles by changing the pK_a value of a catalytic base in an enzyme has been observed previously (31–33). The pK_a value of the active site histidine was raised by 0.2 pH unit by acetylation of all surface lysines in trypsin (31). Shifted pH activity profiles and higher catalytic activity were obtained by changing the surface charge of subtilisin using site-directed mutagenesis (32). The pK_a value of His-64 in the active site of subtilisin was lowered by 0.4 unit at low ionic strength in the range of 0.005–0.01 M by the removal of one surface carboxylate (33). The lowering of the pK_a by employing the change of the surface charge is dependent basically on the long range macroscopic electrostatic interaction, which requires a specific condition of low ionic strength to unmask electrostatic interaction (32). During evolution, enzymes may utilize a long-range electrostatic interaction to regulate the pK_a value of the catalytic residue. However, the low ionic condition to unmask the electrostatic interaction may not be allowed in the biological system. In both intracellular and extracellular environments of a typical mammalian cell, the ion concentration can be higher than 0.1 M (34), and total ion concentration of the bacterial cell was reported to be comparable to that of the mammalian cell (35). The direct regulation by use of the neighboring residue could be more effective than the indirect regulation through the long range electrostatic interaction in the aspect that it is relatively independent of ionic strength of the solution. In this study, we demonstrated that the alteration of the pK_a value of the catalytic residue in a mutant with higher activity than the wild-type enzyme at low pH could be achieved not by changing surface charge but by directly changing the microscopic active-site environment of the catalytic residue utilizing a single hydrogen bond.

In summary, based on comparative studies using two homologous KSIs on the role of a tryptophan, we demonstrated that

tryptophan could alter the pH activity profile by both forming the hydrogen bond with the catalytic base and increasing simultaneously the substrate affinity. PI-WT could employ a kind of a protective strategy to stabilize the deprotonation state of a catalytic base, which can play a critical role as a base catalyst at low pH. This strategy is based on the fine regulation of pK_a of the catalytic base with minimizing a loss of the catalytic activity by increasing the substrate affinity. The enzyme can adapt to the environment to maintain catalytic activity at low pH, independent of external biological buffer during evolution. We assume that this strategy can be found in other enzymes and applied to the alteration of pH activity profile by protein engineering.

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